

# DEVELOPMENT OF A MULTI-CLASS FACIAL EMOTION RECOGNITION ON IMAGES

OMAGE MICHEAL BRIGHT

233353

SUPERVISOR NAME:

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PRESENTATION:

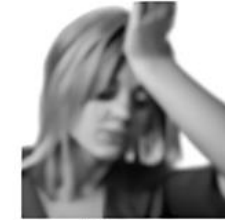
RESULT

# INTRODUCTION

- Facial expressions serve as one of the most natural and powerful means of human communication, capable of conveying a wide spectrum of emotions.
- In the realm of artificial intelligence and computer vision, recognizing these emotions from facial cues has gained significant importance.
- This presentation delves into the fascinating field of development of facial emotion recognition for multi classification of image, leveraging the capabilities of Convolutional Neural Network (CNN).



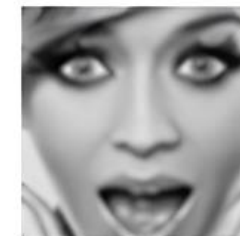
Surprise  
Happiness



Fear  
Sadness



Surprise  
Happiness



Happiness  
Surprise



Fear  
Anger



Sadness  
Fear



Anger  
Sadness



Anger  
Disgust

| Literature Review Table |                     |                                                              |                                                                                                                                              |                                                                                                         |                                                                                    |
|-------------------------|---------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| S/N                     | Authors & Years     | Problem Statement                                            | Methodology                                                                                                                                  | Datasets/Results                                                                                        | Limitation                                                                         |
| 1                       | Kim et al. (2023)   | Difficulty recognizing subtle and blended emotions.          | Proposed multi-output CNN architecture for subtle and blended emotion recognition. Utilized a multi-label approach to handle mixed emotions. | Achieved 67.2% avg F1-score on RAF-DB.                                                                  | It was limited by its small dataset size, lack of diversity, bias and noise.       |
| 2                       | Lee et al. (2022)   | Limits in handling multiple emotion labels.                  | Used an ensemble of CNNs for multi-label prediction                                                                                          | Obtained 58.4% micro F1-score on FERPlus, outperforming single-model methods.                           | Prone to overfitting on small datasets with limited diversity.                     |
| 3                       | Ahmed et al. (2021) | Existing methods do not effectively model multiple emotions. | Proposed a multi-task CNN architecture that jointly learns facial action units and emotion labels                                            | Achieved 52.6% macro F1 on EmotiW 2020, demonstrating improved performance in handling label ambiguity. | Further improvements needed to better handle label ambiguity and blended emotions. |

LITERATURE REVIEW TABLE

|    |                          |                                                                                 |                                                                |                                                                           |                                                       |
|----|--------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------|
| 4  | Wang et al. (2020)       | Limitations in recognizing subtle facial expressions.                           | Used 3D CNNs and attention mechanism.                          | Obtained 73.2% accuracy on CASME II dataset.                              | Tested only on micro-expressions.                     |
| 5  | Liu et al. (2019)        | Handling ambiguities in multi-label emotion data.                               | Proposed CNN-RNN model with joint-loss training.               | Achieved 63.7% F1 score on FERPlus dataset.                               | Did not explore very deep architectures.              |
| 6  | Jung et al. (2017)       | Difficulty recognizing mixtures of facial expressions.                          | Used ensemble of CNNs trained on expression zones.             | Obtained 71.3% accuracy on CK+ dataset.                                   | Limited diversity in dataset.                         |
| 7  | Liu et al. (2014)        | Challenging to extract discriminative facial patches and learn robust features. | Proposed Boosted Deep Belief Network using facial landmarks.   | CK+, MMI, and FERG datasets. Achieved over 90% accuracy.                  | Computationally expensive feature extraction process. |
| 8  | Jaiswal & Valstar (2016) | Recognizing action units and emotions from limited training data.               | Used Bi-directional LSTM neural networks with 3D CNN features. | BP4D, DISFA, and SEMAINE datasets. Obtained 62% average F1 score.         | Did not focus on detecting specific emotions.         |
| 9  | Jung et al. (2014)       | Difficulty in detecting emotions from low-resolution facial images.             | Used deep network combining CNN and RNN with boosting.         | Multi-PIE dataset. Achieved over 80% accuracy from low-resolution images. | Only evaluated on lab-controlled dataset.             |
| 10 | Liu et al. (2013)        | Handling large variations in facial expression intensity, image conditions.     | Proposed 3D CNN combined with deformable model fitting.        | BU-3DFE dataset. Obtained over 85% average accuracy.                      | Computationally intensive 3D modeling                 |

# PROBLEM STATEMENT

Current Facial Expression Recognition (FER) systems primarily focus on detecting single dominant emotions and struggle with multi-classifying an image based on probability and chances an image might have more than one emotion. This limitation highlights the pressing need for more sophisticated approaches capable of accurately modeling emotion mixtures.

The task of recognizing facial expressions is inherently challenging. It becomes particularly intricate when faced with blended expressions, where multiple emotions coexist in a single facial display. Additionally, detecting subtle emotions, like faint smiles, demands a precise approach. Addressing these complexities necessitates the application of advanced techniques, and Convolutional Neural Network emerge as a promising solution.

# AIM AND OBJECTIVES

Aim of the project:

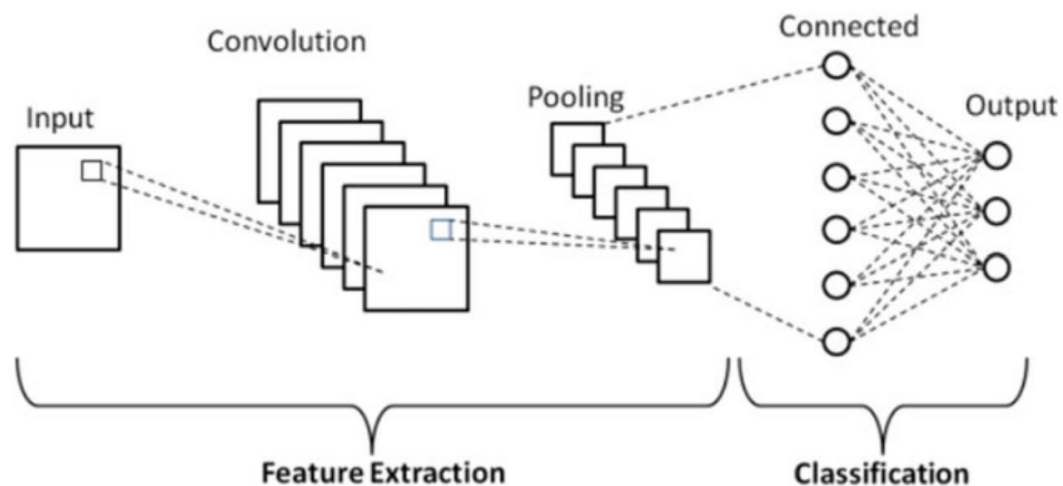
Develop a FER system that can predict and quantify the presence of multiple emotions in a single facial image, providing a list of emotions along with their respective probabilities.

## SPECIFIC OBJECTIVES

1. Design CNN architecture optimized for probability prediction on 8 classes of facial emotions.
2. Analyze model for blended emotion insights.
3. Evaluate using metrics like micro/macro F1 score, and ROC curve.

# Methodology

## Training Stage



Facial Expression

Dataset



Crop Face

Cleaned  
Dataset

Fine-  
Tuning

Emotion  
Recognition  
Model

## Testing Stage



Test Image

Crop Face



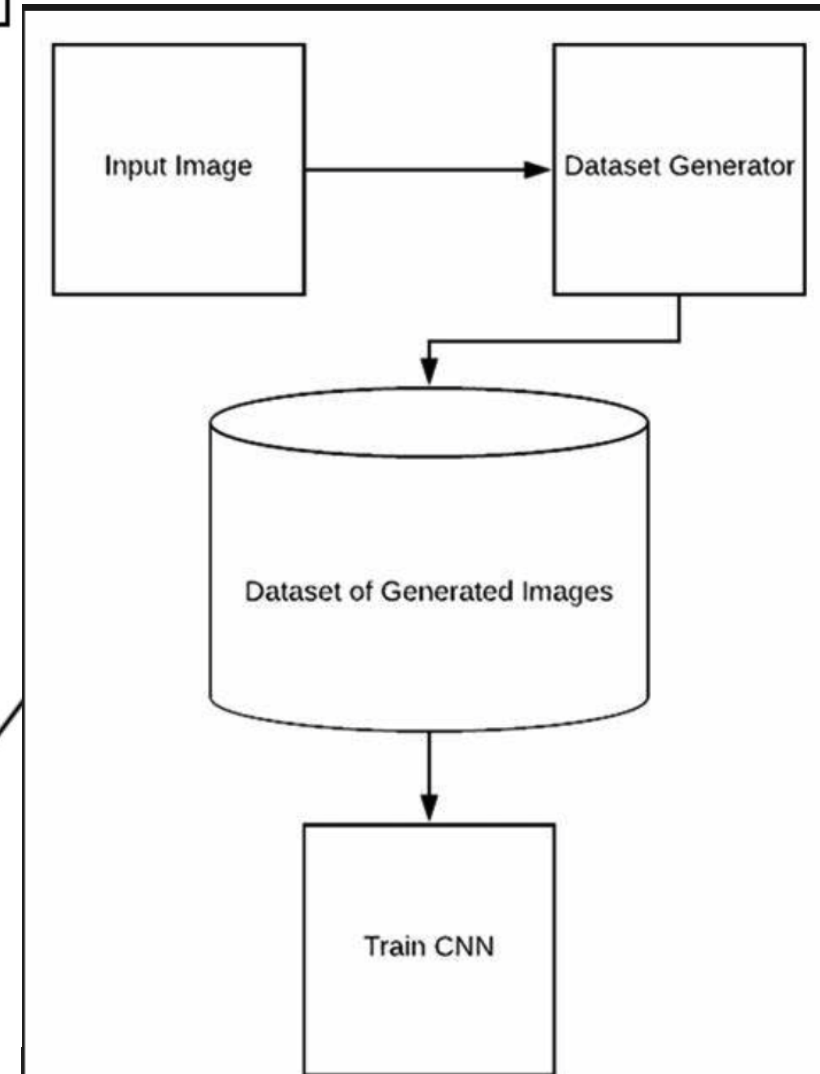
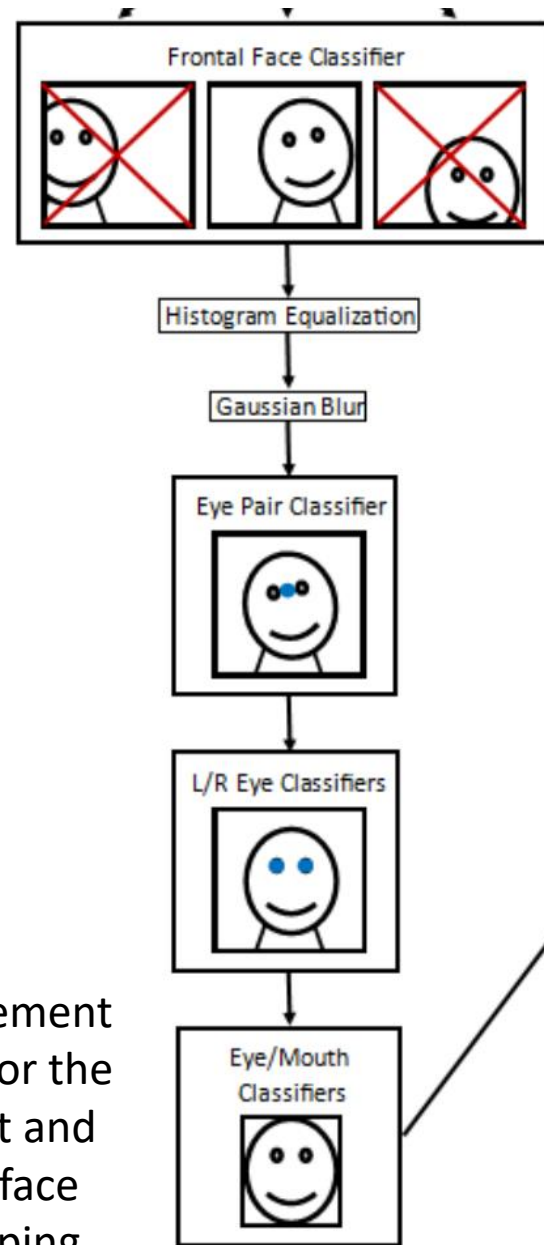
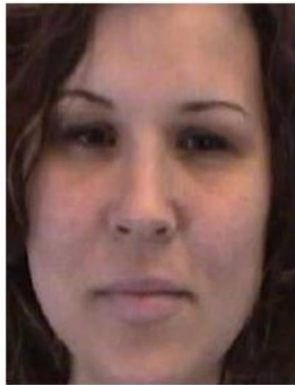
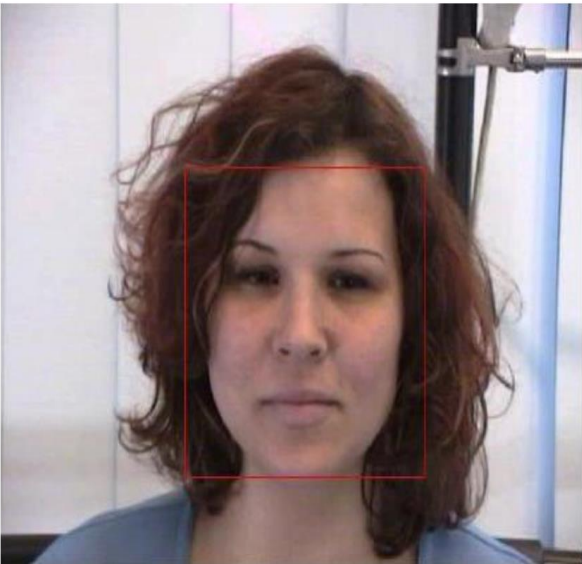
Cropped Face

Emotion  
Recognition  
Model

|           |
|-----------|
| Afraid    |
| Angry     |
| Disgusted |
| Sad       |
| Happy     |
| Surprised |
| Neutral   |

Output  
Probability  
of Each  
Emotion

# Dataset Preprocessing and Argumentation



The use of OpenCV libraries helps in the Capturing of Facial Images by using histogram equalization for contrast enhancement and Gaussian blur for noise reduction, which makes it easy for the detection of eyes using an eye classifier, by detecting the left and right eye, makes it easier for OpenCV to align itself with the face rotation so as to detect the nose and mouth, for better cropping of the image

## More information about Dataset

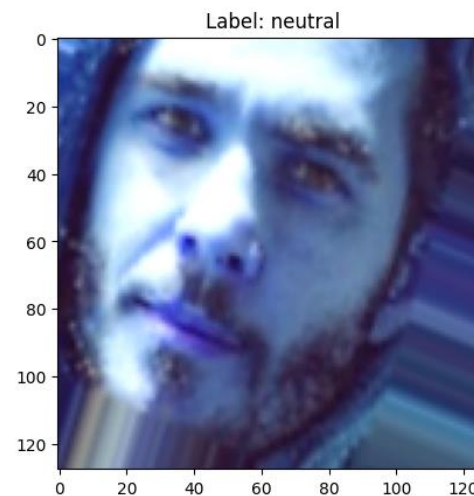
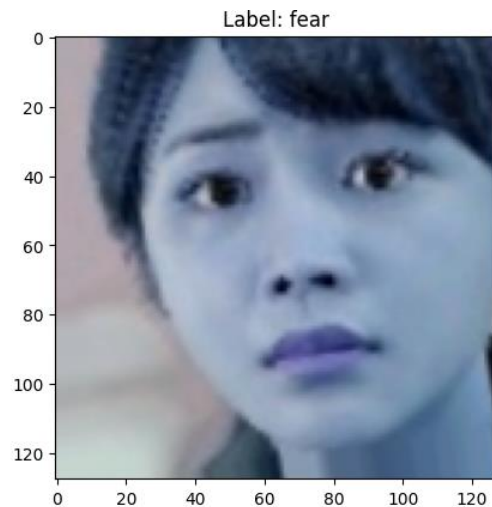
The AffectNet dataset has over 27,836 images with 8 classes of images which are :-

- Anger
- Contempt
- Disgust
- Fear
- Happy
- Neural
- Surprise
- Sad

Each images are of size 96 X 96, which are in the form RGB, which is publicly accessible through Kaggle datasets and research publications, AffectNet empowers researchers and developers to leverage its rich database for facial expression analysis and emotion recognition advancements, while I use FER Plus dataset which has over 35,710 images to evaluate the model, although the images are in form of grey scale but maintain at 3 channels

# Data Generator

```
data_generator = ImageDataGenerator(  
    rescale=1./255,  
    width_shift_range=0.1,  
    height_shift_range=0.1,  
    brightness_range=(0.8,1.2),  
    zoom_range=0.1,  
    horizontal_flip=True,  
    vertical_flip=False,  
    rotation_range=20,  
    #fill_mode="bicubic",  
    fill_mode='nearest',  
    data_format='channels_last'  
)
```



```
[[[0.8313726  0.74509805 0.882353 ]  
 [0.8313726  0.74509805 0.882353 ]  
 [0.8313726  0.74509805 0.882353 ]  
 ...  
 [0.6627451  0.43137258 0.81568635]  
 [0.6862745  0.4431373  0.8352942 ]  
 [0.6901961  0.44705886 0.8431373 ]]  
  
[[[0.8313726  0.74509805 0.882353 ]  
 [0.8313726  0.74509805 0.882353 ]  
 [0.8313726  0.74509805 0.882353 ]  
 ...  
 [0.67058825 0.4431373  0.8235295 ]  
 [0.68235296 0.44705886 0.83921576]  
 [0.6745098  0.43921572 0.8431373 ]]
```

## Benefits of Data Augmentation:

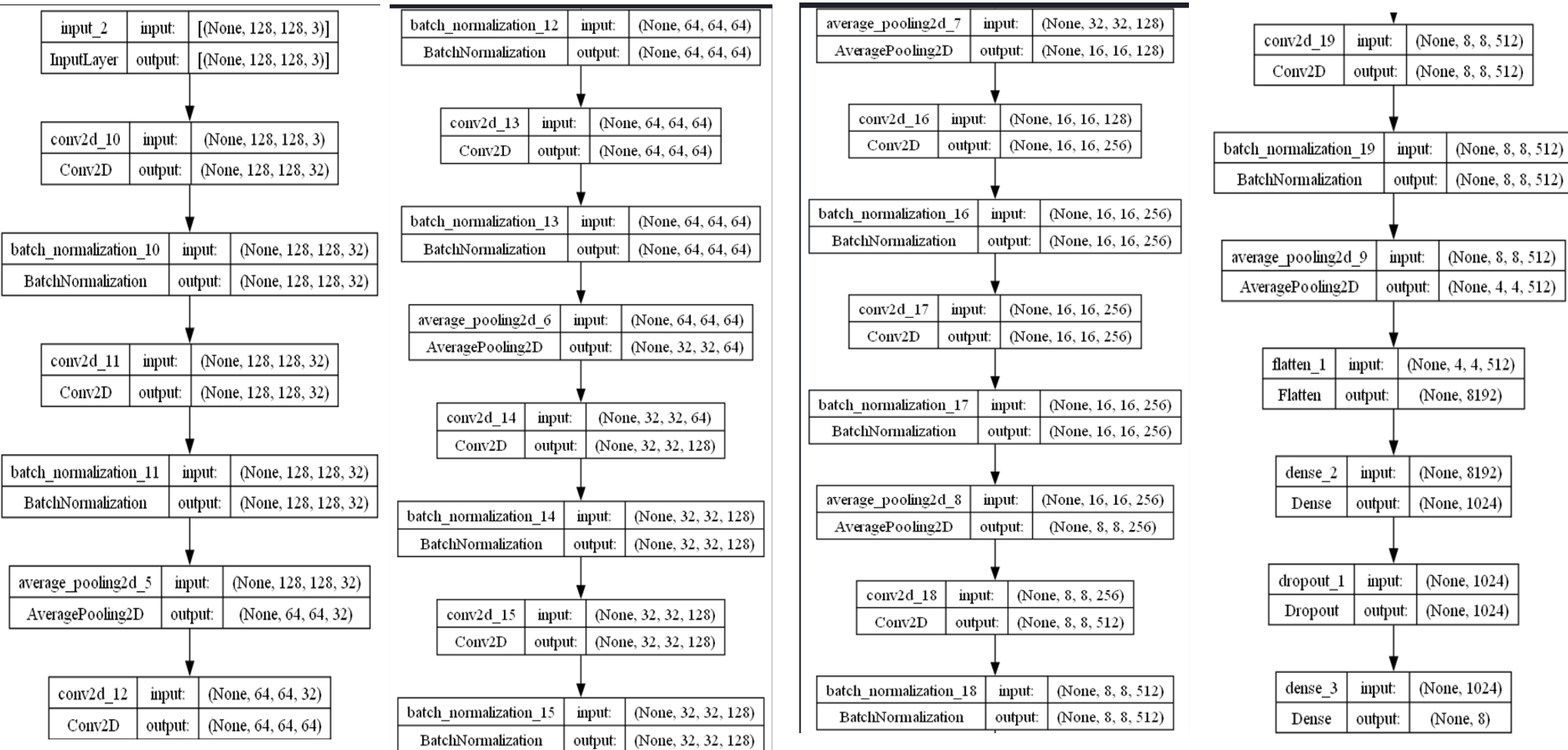
- Prevents Overfitting: By introducing variations, data augmentation helps the model learn features that are relevant to the task at hand, rather than memorizing specific details of the training images. This leads to a model that performs better on unseen data.
- Improves Generalizability: The model becomes more robust to small variations in the data it encounters during deployment.

# Data Generator

- Key Parameters:
- `rescale=1./255`: This line scales the pixel values of your images from a range of [0, 255] (typical for image data) to a range of [0, 1]. This normalization is often a good practice for training image classification models.
- `width_shift_range=0.1, height_shift_range=0.1`: These parameters introduce random horizontal and vertical shifts of the image content within a range of 10% of the image width and height, respectively. This simulates slightly different camera angles or object positions.
- `brightness_range = (0.8,1.2)`: This parameter randomly adjusts the brightness of the images by a scale factor between 0.8 (darker) and 1.2 (brighter). This helps the model learn to recognize objects under different lighting conditions.
- `zoom_range=0.1`: This parameter applies random zoom of up to 10% to the images. This simulates objects being slightly closer or farther away from the camera.
- `horizontal_flip=True, vertical_flip=False`: This setting performs a random horizontal flip (mirroring) of the image with a probability of True. Vertical flipping is set to False here. This introduces variations in object orientation.
- `rotation_range = 20`: This parameter applies random rotations to the images within a range of 20 degrees. This helps the model recognize objects from different viewpoints.
- `fill_mode='nearest'`: This parameter specifies how to fill the background pixels that become exposed due to random shifts, zooms, or rotations.
- `data_format='channels_last'`: This parameter specifies the data format expected by the model. Here, 'channels\_last' indicates that color channels are stored at the end of each pixel value (e.g., [red, green, blue]).

# CNN Architecture: Objective 1

Total params: 13,117,992  
Trainable params: 13,114,024  
Non-trainable params: 3,968



# CNN Training Result Analysis

## #Building of Model

```
i = Input(shape=(128,128, 3))
x = Conv2D(32,(3,3), activation = "relu", padding = "same")(i)
x = BatchNormalization()(x)

x = Conv2D(32,(3,3), activation = "relu", padding = "same")(x)
x = BatchNormalization()(x)
x = AveragePooling2D((2,2))(x)

x = Conv2D(64,(3,3), activation = "relu", padding = "same")(x)
x = BatchNormalization()(x)

x = Conv2D(64,(3,3), activation = "relu", padding = "same")(x)
x = BatchNormalization()(x)
x = AveragePooling2D((2,2))(x)

x = Conv2D(128,(3,3), activation = "relu", padding = "same")(x)
x = BatchNormalization()(x)

x = Conv2D(128,(3,3), activation = "relu", padding = "same")(x)
x = BatchNormalization()(x)
x = AveragePooling2D((2,2))(x)

x = Conv2D(256,(3,3), activation = "relu", padding = "same")(x)
x = BatchNormalization()(x)

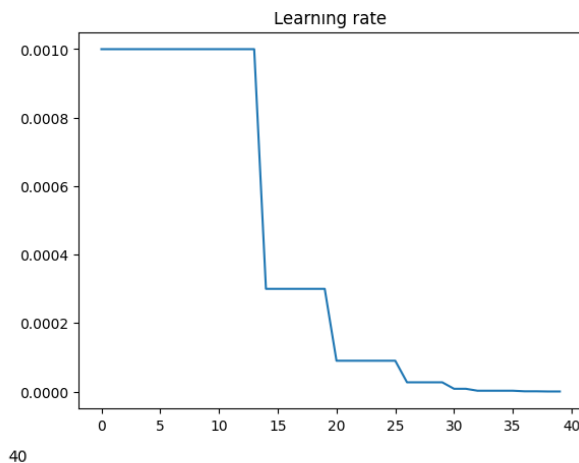
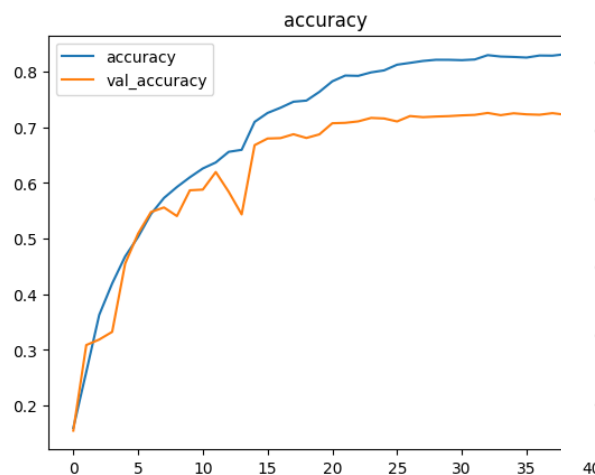
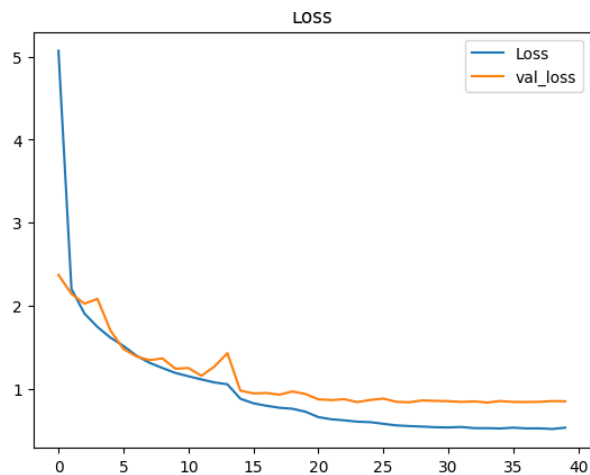
x = Conv2D(256,(3,3), activation = "relu", padding = "same")(x)
x = BatchNormalization()(x)
x = AveragePooling2D((2,2))(x)

x = Conv2D(512,(3,3), activation = "relu", padding = "same")(x)
x = BatchNormalization()(x)

x = Conv2D(512,(3,3), activation = "relu", padding = "same")(x)
x = BatchNormalization()(x)
x = AveragePooling2D((2,2))(x)

x = Flatten()(x)
#x = Dropout(0.3)(x)
x = Dense(1024, activation='relu', kernel_regularizer='l2')(x)
x = Dropout(0.2)(x)
x = Dense(8, activation='softmax')(x) # 8 emotion classes
```

```
model = Model(i,x)
```



```
Epoch 9/40
733/733 [=====] - 62s 84ms/step - loss: 1.2536 - accuracy: 0.5926 - val_loss: 1.3692 - val_accuracy: 0.5404 - lr: 0.0010
Epoch 10/40
733/733 [=====] - 93s 127ms/step - loss: 1.1947 - accuracy: 0.6101 - val_loss: 1.2454 - val_accuracy: 0.5867 - lr: 0.0010
Epoch 11/40
733/733 [=====] - 112s 153ms/step - loss: 1.1547 - accuracy: 0.6259 - val_loss: 1.2537 - val_accuracy: 0.5879 - lr: 0.0010
Epoch 12/40
733/733 [=====] - 103s 140ms/step - loss: 1.1165 - accuracy: 0.6367 - val_loss: 1.1607 - val_accuracy: 0.6195 - lr: 0.0010
Epoch 13/40
...
Epoch 40/40
733/733 [=====] - ETA: 0s - loss: 0.5360 - accuracy: 0.8261
Epoch 40: ReduceLROnPlateau reducing learning rate to 6.561000276406048e-08.
733/733 [=====] - 72s 98ms/step - loss: 0.5360 - accuracy: 0.8261 - val_loss: 0.8533 - val_accuracy: 0.7194 - lr: 2.1870e-07
```

# Analyze model for blended emotion insights: Objective 2

['anger', 'contempt', 'disgust', 'fear', 'happy', 'neutral', 'sad', 'surprise']

Real Image from <https://unsplash.com/>



[[3.0024806e-01 7.2210595e-02 7.3986992e-02 1.7576167e-02  
1.5977972e-04 8.0852937e-03 5.2031827e-01 7.4148248e-03]]

[[0.3 0.072 0.074 0.018 0. 0.008 0.52 0.007]]

['anger', 'contempt', 'disgust', 'fear', 'happy', 'neutral', 'sad', 'surprise']

DALL-E 3 Model



[[2.6173051e-02 7.1953731e-03 2.1232111e-02 3.8065571e-01  
9.5840449e-05 1.3002047e-03 5.4262662e-01 2.0721048e-02]]

[[0.026 0.007 0.021 0.381 0. 0.001 0.543 0.021]]

['anger', 'contempt', 'disgust', 'fear', 'happy', 'neutral', 'sad', 'surprise']

a face of a boy that is sad and fearful

Cocreator

PREVIEW

Describe what you'd like to create

a face of a boy that is sad and fearful

Choose a style

Photorealistic

Explore variants

43

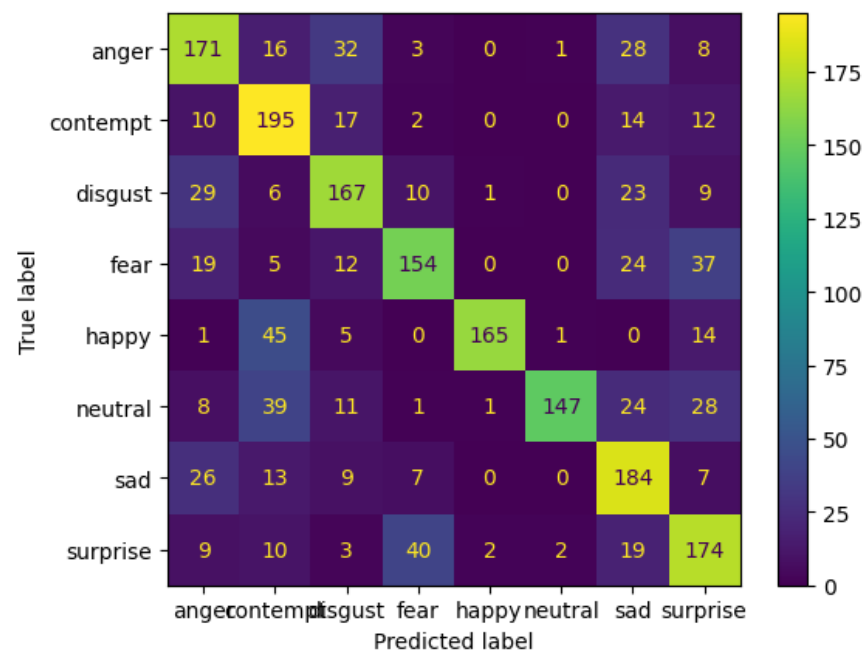
Create

# Performance and Evaluation: Objective 3

- Classification Report and Confusion Metrics:

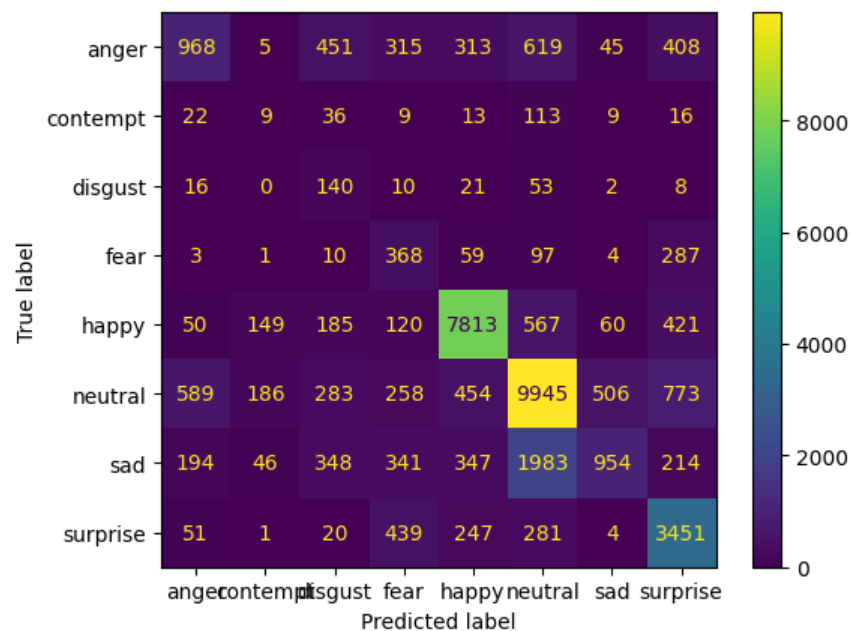
- Test Set

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| anger        | 0.63      | 0.66   | 0.64     | 259     |
| contempt     | 0.59      | 0.78   | 0.67     | 250     |
| disgust      | 0.65      | 0.68   | 0.67     | 245     |
| fear         | 0.71      | 0.61   | 0.66     | 251     |
| happy        | 0.98      | 0.71   | 0.82     | 231     |
| neutral      | 0.97      | 0.57   | 0.72     | 259     |
| sad          | 0.58      | 0.75   | 0.65     | 246     |
| surprise     | 0.60      | 0.67   | 0.64     | 259     |
| accuracy     |           |        | 0.68     | 2000    |
| macro avg    | 0.71      | 0.68   | 0.68     | 2000    |
| weighted avg | 0.71      | 0.68   | 0.68     | 2000    |



- FER-Plus Dataset for Evaluation

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| anger        | 0.51      | 0.31   | 0.39     | 3124    |
| contempt     | 0.02      | 0.04   | 0.03     | 227     |
| disgust      | 0.10      | 0.56   | 0.16     | 250     |
| fear         | 0.20      | 0.44   | 0.27     | 829     |
| happy        | 0.84      | 0.83   | 0.84     | 9365    |
| neutral      | 0.73      | 0.77   | 0.75     | 12994   |
| sad          | 0.60      | 0.22   | 0.32     | 4427    |
| surprise     | 0.62      | 0.77   | 0.69     | 4494    |
| accuracy     |           |        | 0.66     | 35710   |
| macro avg    | 0.45      | 0.49   | 0.43     | 35710   |
| weighted avg | 0.69      | 0.66   | 0.66     | 35710   |



# Performance and Evaluation

- Performance Metrics:

Micro/Macro F1-score:

F1-score is a harmonic mean of precision and recall, balancing both measures for a more robust evaluation.

$$F1 = 2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall})$$

Micro F1-score: Calculates the overall average F1-score across all classes, potentially favoring dominant classes.

Equation:

$$\text{Micro F1} = (\text{sum of true positives across all classes}) / (\text{sum of predicted positives across all classes})$$

Macro F1-score: Averages the individual F1-scores for each class, giving equal weight to all emotions.

Equation:

$$\text{Macro F1} = (\text{sum of individual class F1-scores}) / \text{number of classes}$$

Test Set

Micro F1 Score: 0.6785

Macro F1 Score: 0.6841415668402315

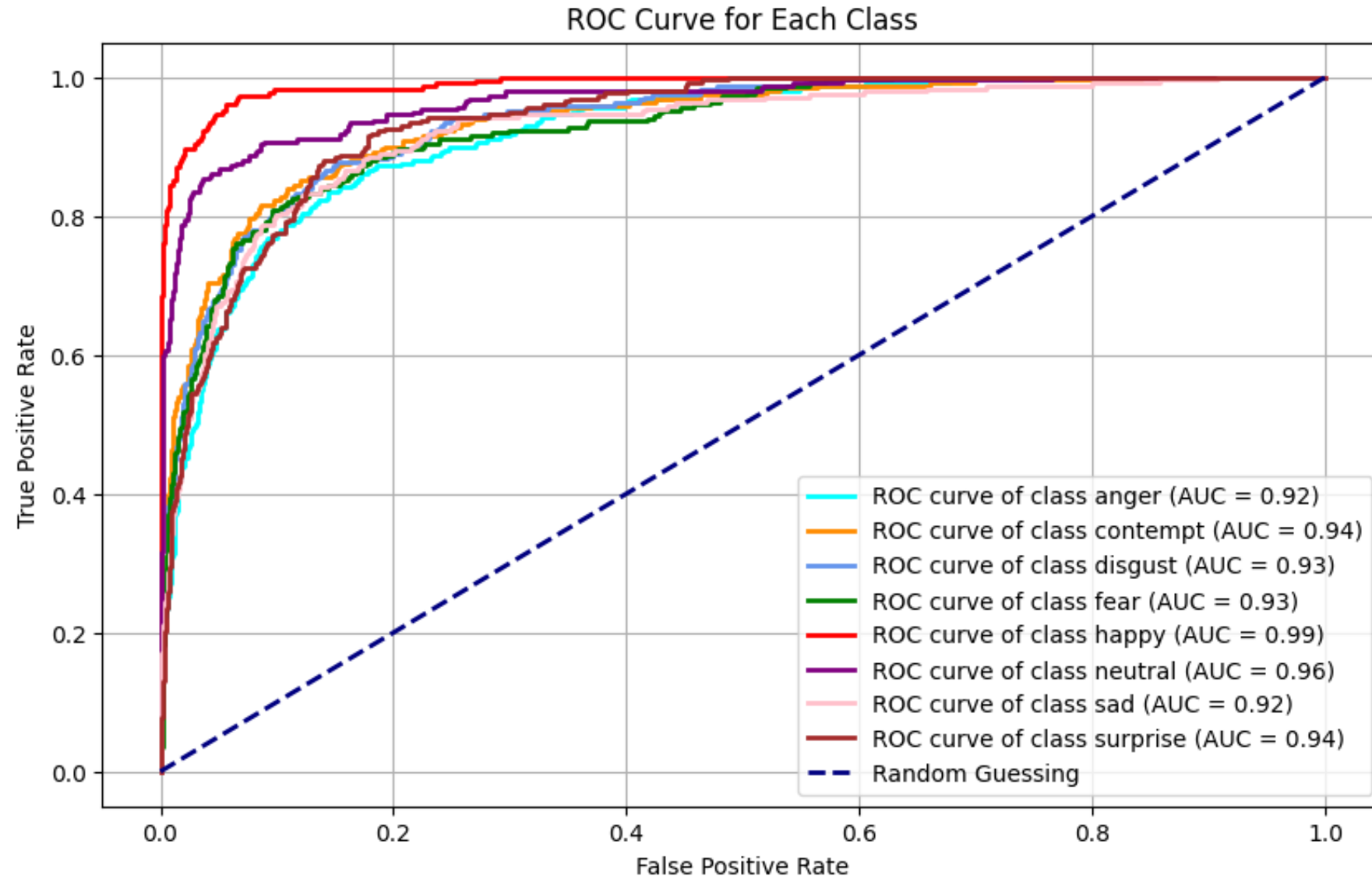
FER-Plus Dataset for Evaluation

Micro F1 Score: 0.6623074768972277

Macro F1 Score: 0.42990340277361966

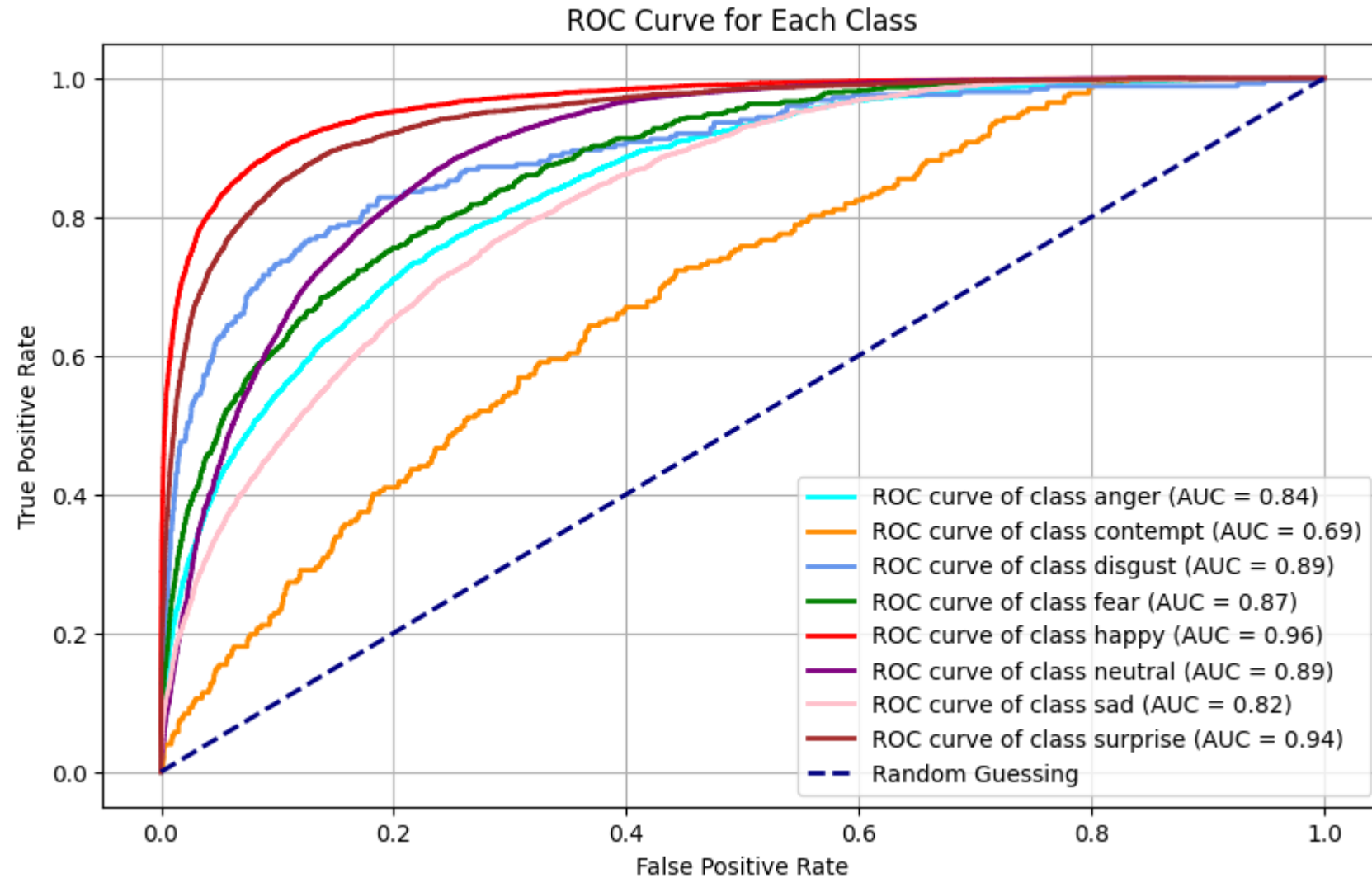
# Evaluation with ROC Curve: Objective 3

- Test Set



# Evaluation with ROC Curve: Objective 3

- FER Plus Dataset



# EXPECTED CONTRIBUTION TO KNOWLEDGE

- Advance state-of-the-art in multi-label FER.
- Provide insights into blended facial emotion recognition.
- Enable new applications benefitting from facial emotion detection using probability.

# REFERENCES

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